

circumstances such as job loss or illness can reduce it. The riskiness of a person's labor income—its volatility and how it correlates with financial returns—affects her asset allocation.

We take account of human capital by treating it like an asset.¹⁰ Suppose we denote the present value of an individual's wages as H (for human capital). Human capital is the discounted value of a person's lifelong earnings from work; it is the present value today that is equivalent to the wage profile of that individual. Human capital is risky because earnings from work can fluctuate—and in the most unfortunate case fall to and stay at zero.

We denote financial wealth as W . Total wealth is $H+W$, or, in other words, total wealth comprises both human capital and financial wealth.

2.1. ASSET ALLOCATION WITH NO LABOR INCOME

Consider an individual with no labor income and \$1 million of financial assets. For simplicity, we'll assume that there are only equities, which have an expected return of $\mu = 10\%$ and a volatility of $\sigma = 15\%$, and risk-free T-bills paying $r_f = 2\%$. The investor has mean-variance utility, or equivalently has CRRA utility, with a risk aversion of $\gamma = 5$. (Chapters 2 and 3 cover utility functions and mean-variance investing, respectively.¹¹)

Our investor with only financial wealth holds $w^* = 62.5\%$ in equities, which is obtained from the mean-variance investing formula:

$$w^* = \frac{1}{\gamma} \frac{\mu - r_f}{\sigma^2} = \frac{1}{5} \frac{0.10 - 0.02}{(0.15)^2} = 62.5\% \quad (5.1)$$

(See equation (2.10) in chapter 2.) His asset allocation is then:

Hold $0.625 \times \$1 \text{ million} = \$625,000$ in equities
<u>\$375,000 in T-bills</u>
\$1 million

2.2. ASSET ALLOCATION WITH RISK-FREE LABOR INCOME

Now contrast the asset allocation of an individual who has risk-free wages, like tenured professors at top universities (yes, I know, I'm lucky and very grateful) or federal judges. Her human capital is equivalent to her being endowed with risk-free assets. Let's assume her wage flow is equivalent to human capital today of

¹⁰ This approach follows Merton (1990) and Bodie, Merton, and Samuelson (1992).

¹¹ In this example, we take risk aversion and the amount of human capital wealth (and total wealth) as given. In reality, risk aversion affects both the choice of financial assets and the choice and type of human capital—a highly risk-averse person is unlikely to pursue a career as a mountaineer, for example. See Ranish (2012) and section 3.6 on endogenous labor income.